IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of Methodologies**
- Data collection using SpaceX REST API and web scraping
- Data cleaning, handling missing values, and feature encoding
- EDA using SQL queries and data visualization
- Interactive analysis using Plotly Dash and Folium maps
- Predictive modeling using classification algorithms with hyperparameter tuning

- **Summary of All Results**
- Launch success is strongly influenced by **payload mass, launch site, and orbit type**
- **KSC LC-39A** showed the highest success rate among launch sites
- Lower payload ranges achieved higher landing success
- The **Decision Tree model** achieved the best prediction accuracy after tuning
- Results confirm that historical launch data can effectively predict landing outcomes

Introduction

- **Project Background and Context**

- SpaceX has introduced **reusable Falcon 9 rockets**, depicting significant reduction in launch costs compared to traditional expendable rockets.
- The major contributing factors are the **successful landing and reuse of the first stage booster**. But not all launches result in a successful landing.
- Understanding the factors that influence **first stage landing success** is essential for estimating launch costs and evaluating SpaceX's competitive advantage.
- By analyzing historical launch data, data science techniques can be applied to identify patterns and develop predictive models that support informed decision-making in the commercial space sector.

Introduction

- Problems to find answers through the Project

1. Can the success of a Falcon 9 first stage landing be **predicted using historical launch data**?
2. Which factors (such as **payload mass, orbit type, and launch site**) have the greatest impact on landing success?
3. How does landing success vary across **different launch sites and mission profiles**?
4. Which **machine learning classification model** provides the most accurate prediction of landing success?

Section 1

Methodology

Methodology

Executive Summary

Data Collection Methodology

- Data was collected using the **SpaceX REST API** to obtain structured information on Falcon 9 launches, including payload details, orbit type, launch site, and landing outcomes. Additional historical launch records were gathered through **web scraping from Wikipedia** to ensure data completeness.
- **Data Wrangling**
- The collected data was cleaned by identifying and handling missing or inconsistent values. Categorical variables such as launch site and orbit type were converted into numerical form using **one-hot encoding**, and a binary target variable (**Class**) was created to represent landing success or failure.
- **Exploratory Data Analysis (EDA)**
- EDA was performed using **SQL queries** and **data visualization techniques** to analyze launch sites, payload mass, mission outcomes, and success rates. Visualizations helped identify trends, relationships, and key factors influencing landing success.

Methodology

Executive Summary

- **Interactive Visual Analytics**
- Interactive visual analysis was conducted using **Folium** and **Plotly Dash**. Folium maps were used to visualize launch locations and geographical patterns, while the Dash dashboard enabled dynamic exploration of payload mass and landing outcomes through interactive charts.
- **Predictive Analysis**
- Several **classification models** were developed to predict first stage landing success, including Logistic Regression, SVM, KNN, and Decision Tree. These models learned patterns from historical launch data.
- **Model Building, Tuning, and Evaluation**
- Models were trained using a train-test split and tuned with **GridSearchCV** to identify optimal hyperparameters. Performance was evaluated using **accuracy scores and confusion matrices**, and the **Decision Tree model** achieved the best overall performance.

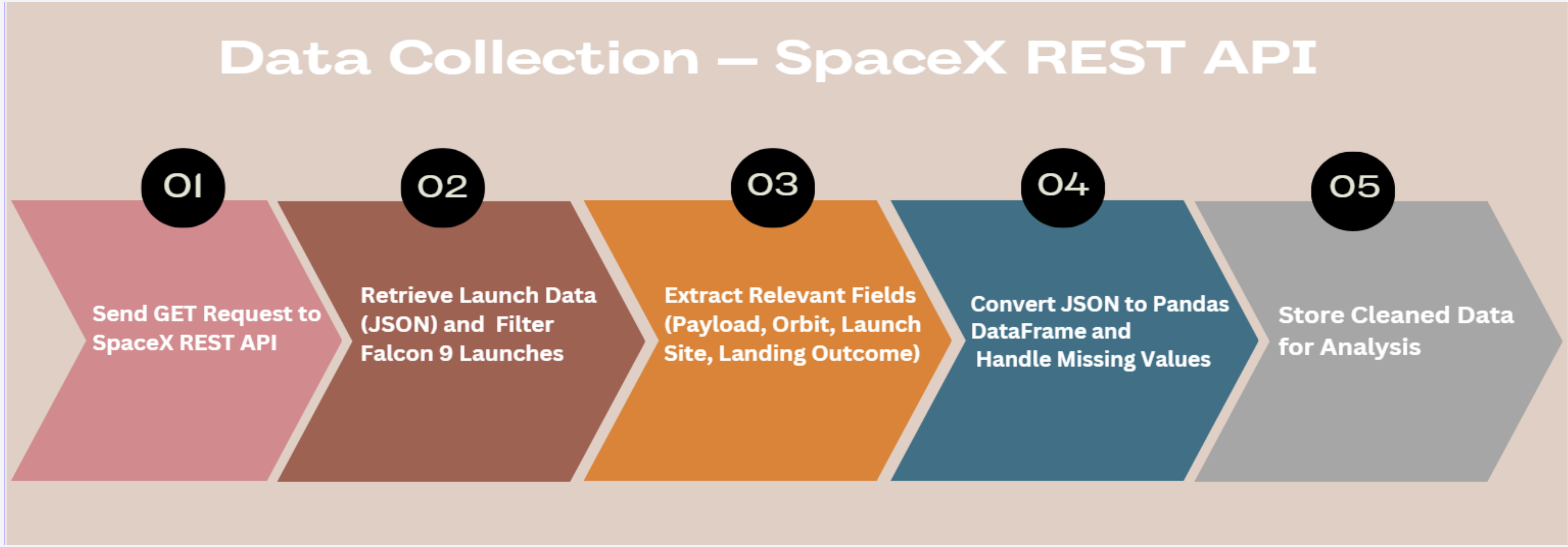
Data Collection

- Data was collected using the SpaceX REST API to obtain detailed launch and rocket information.
- The SpaceX REST API provides structured and reliable launch data, which allows efficient extraction, filtering, and preparation of Falcon 9 mission records for further analysis.
- Additional historical launch data was gathered through web scraping from Wikipedia to ensure completeness of Falcon 9 launch records.

Data Collection – SpaceX API

- Data was collected using the SpaceX REST API (v4).
- HTTP GET requests were used to retrieve launch and rocket information.
- API endpoints provided details such as:
  - Launch date and site
  - Rocket and booster information
  - Payload mass and orbit type
  - First stage landing outcome
- Data was filtered to include Falcon 9 launches only.
- API responses were returned in JSON format and converted into structured datasets using Pandas for analysis.

Data Collection – SpaceX API

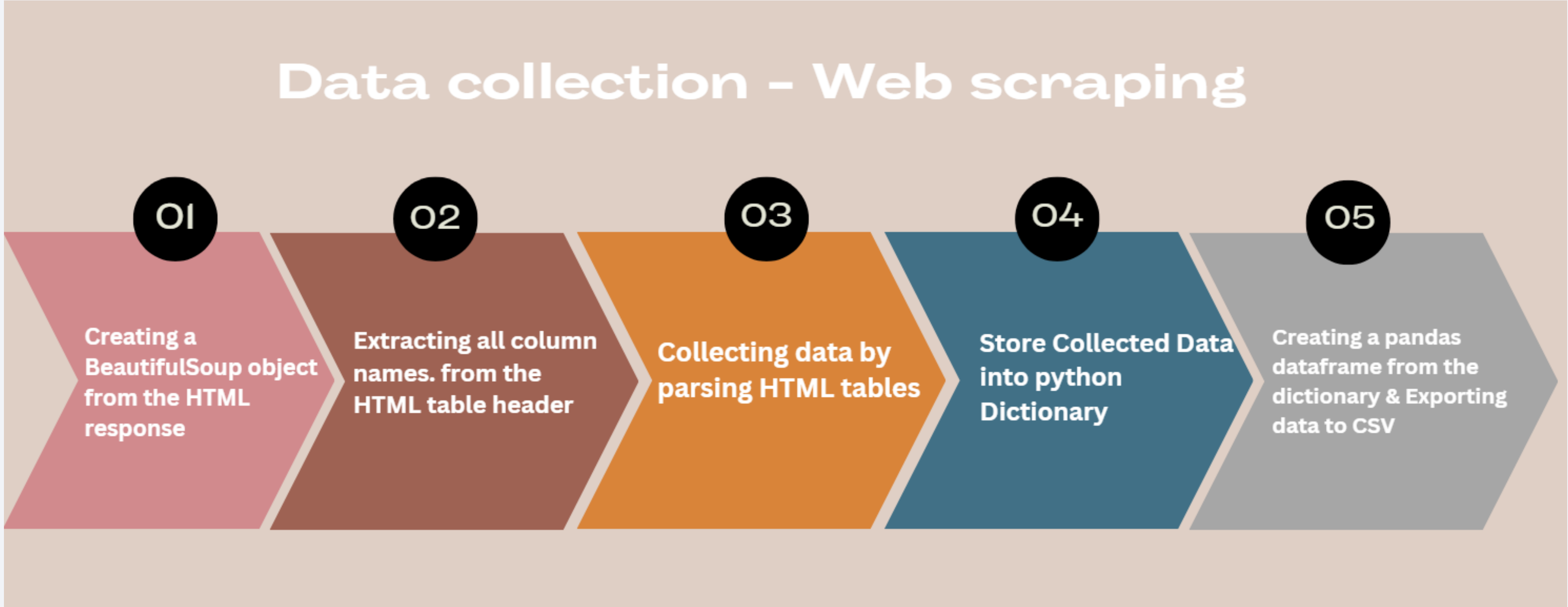


- https://github.com/Hardika13/IBMDS-CapstoneProject/blob/main/IBM%20DS%20CapstoneLab1-spacex-data-collection-api.ipynb

Data Collection - Scraping

- Historical Falcon 9 launch data was collected through **web scraping from Wikipedia**.
- The **Requests** library was used to retrieve the web page content.
- **BeautifulSoup** was applied to parse HTML tables containing launch records.
- Relevant fields such as **launch date, booster version, payload mass, orbit type, and landing outcome** were extracted.
- Unnecessary columns were removed, and data formats were standardized.
- The scraped data was converted into a **Pandas DataFrame** for further cleaning and analysis.

Data Collection - Scraping



- https://github.com/Hardika13/IBMDS-CapstoneProject/blob/main/IBM_DS_CapstoneLab2_webscraping_.ipynb

Data Wrangling

- Initially missing and inconsistent values in dataset were identified to ensure data quality and reliability. Numerical missing values, such as payload mass, were handled using appropriate statistical methods (for example, replacing with the mean), while incomplete or irrelevant records were removed.
- Categorical variables, including **launch site**, **orbit type**, and **booster version**, were transformed into numerical format using **one-hot encoding, as machine learning model require numeric values**.
- Finally, a target variable named **Class** was created for the landing outcome of Falcon 9 first stage. A value of **1** indicates a successful landing, while **0** represents unsuccessful/failed landing. This will be used by Supervised learning models.
- https://github.com/Hardika13/IBMDS-CapstoneProject/blob/main/IBM_DS_Capstone Lab3_spacex_Data_wrangling_v2.ipynb

EDA with Data Visualization

- Data visualization techniques were applied to transform dense numbers into easy-to-understand visuals.
- Scatter plots, bar charts, and line charts were used to analyze how **payload mass varies with landing success**, how **orbit type impacts success rate**, and how **launch sites differ in performance over time**.
- Visual analysis made it easier to identify trends, correlations, and outliers that are not easily observed through numerical analysis alone, which helps in better feature selection for predictive modeling.
- https://github.com/Hardika13/IBMDS-CapstoneProject/blob/main/IBM_DS_CapstoneLab4_EDA_SQL.ipynb

EDA with SQL

- Structured Query Language (SQL) was used to explore and summarize the dataset stored in tabular form. SQL queries helped extract meaningful insights by filtering, aggregating, and grouping the data and provided a clear quantitative understanding of how different factors influence mission success.
- Key analyses included identifying **unique launch sites**, calculating **total and average payload mass**, counting **successful and failed mission outcomes**, and analyzing **landing success trends across different launch conditions**.
- These queries https://github.com/Hardika13/IBMDS-CapstoneProject/blob/main/IBM_DS_CapstoneLab5_EDA_Visualization.ipynb

Build an Interactive Map with Folium

- An interactive map was created using the **Folium** library to visualize the geographical locations of SpaceX Falcon 9 launch sites.
- **Markers and color-coded icons** were used to distinguish between successful and failed landing outcomes.
- Circles and lines were added to represent **distances between launch sites and nearby infrastructure**, such as coastlines, railways, and cities.
- The map helps identify **geographical patterns** and environmental factors that may influence landing success.
- https://github.com/Hardika13/IBMDS-CapstoneProject/blob/main/IBM_DS_CapstoneLab6_InteractiveFolium.ipynb

Build a Dashboard with Plotly Dash

- An interactive dashboard was developed using **Plotly Dash**. The dashboard allows users to interact with the data through **dropdown menus and range sliders**, making it easier to analyze landing success under different launch conditions.
- The dashboard includes a **pie chart** that displays the proportion of successful and failed landings for each launch site, allowing quick comparison of performance across locations.
- A **scatter plot** was also created to visualize the relationship between **payload mass and landing outcome**, where users can adjust the payload range to observe how mission success varies with payload size.
- Interactive components such as **launch site selection** and **payload range filtering** update the visualizations in real time.
- https://github.com/Hardika13/IBMDS-CapstoneProject/blob/main/IBM_DS_CapstoneLab6_InteractiveFolium.ipynb

Predictive Analysis (Classification)

- Supervised **classification models** were used to predict whether the Falcon 9 first stage would land successfully.
- The dataset was split into **training and testing sets** and 4 supervised classification algorithms were implemented: Logistic Regression, SVM, KNN and Decision Tree.
- Numerical features were **standardized** to improve model performance.
- For hyper parameter tuning of each model, the GridSearchCV was applied
- Models were evaluated using:
  - **Accuracy score**
  - **Confusion matrix**
- Cross-validation results were compared to select the **best-performing model**.

Predictive Analysis (Classification)

Predictive analysis (Classification)

Creating a NumPy array from the column "Class" in data

Standardizing the data with StandardScaler, then fitting and transforming it

Splitting the data into training and testing sets with train test split function

Creating a GridSearchCV object to find best parameters.

z

Examining confusion matrix. And Calculating accuracy for all models

Applying GridSearchCV on LogReg, SVM, Decision Tree, and KNN models

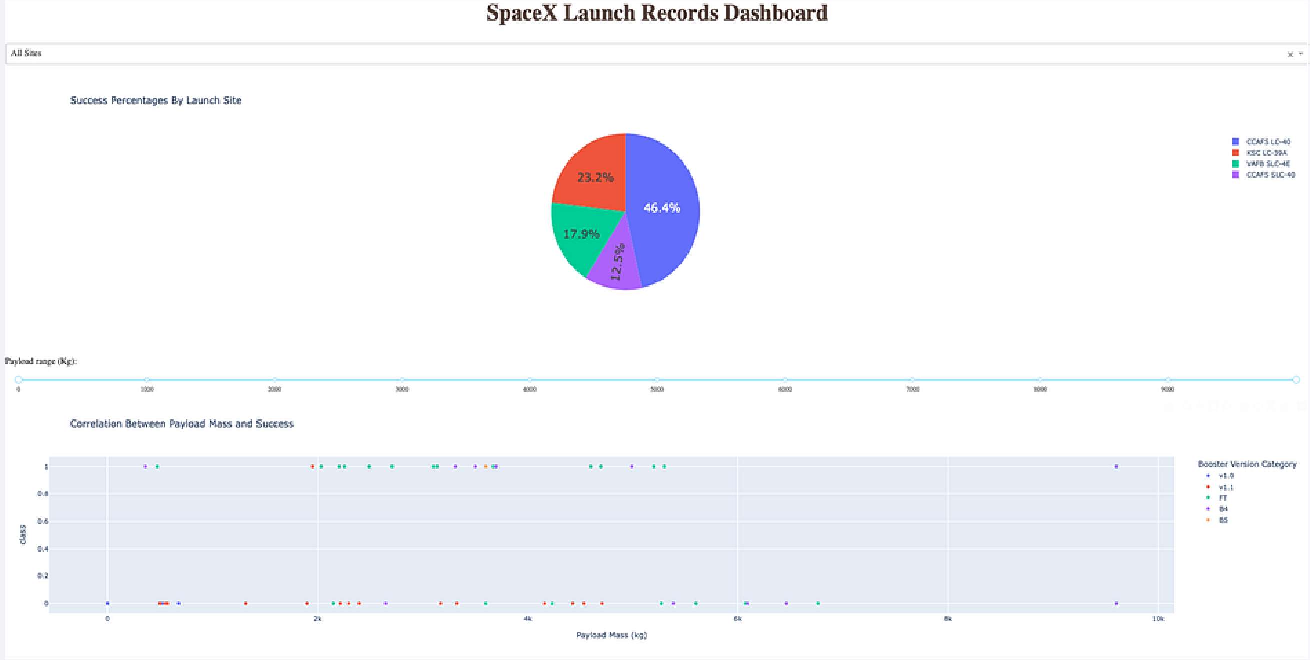
- https://github.com/Hardika13/IBMDS-CapstoneProject/blob/main/IBM_DS_CapstoneLab7_ML_Prediction.ipynb

Results

- **Exploratory Data Analysis Results**
- Analysis revealed that **payload mass and orbit type** have a significant impact on first stage landing success.
- Certain **launch sites** consistently showed higher success rates compared to others.
- Missions with **lighter to medium payloads** generally had a higher probability of successful landings.
- Yearly trend analysis showed an **improvement in landing success over time**, indicating operational learning and optimization by SpaceX.

Results

- **Interactive Analytics Demo (Screenshots)**
- **Plotly Dash dashboard** allows users to interactively explore launch outcomes and gives deeper insights compared to static visualizations.
- **Pie charts** showing success vs failure by launch site. **Scatter plots** display the relationship between payload mass and landing outcome, with interactive filters.



Results

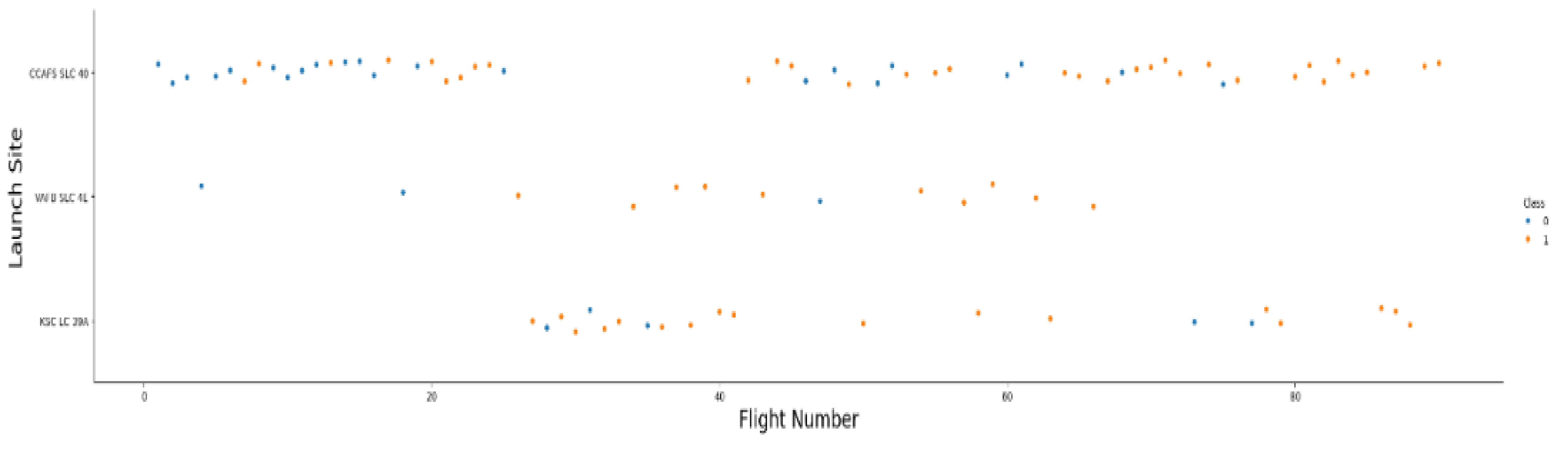
- **Predictive Analysis Results**
- Multiple classification models were evaluated, including **Logistic Regression, SVM, KNN, and Decision Tree**.
- All models achieved comparable test accuracy; however, **Decision Tree** performed best based on cross-validation scores.
- After hyperparameter tuning using **GridSearchCV**, the optimal configuration helped prevent overfitting while preserving important decision rules.
- The confusion matrix further supports the model's strong performance, showing a **balanced number of correct predictions** for both successful and failed landings.
- This indicates that the model did not favor one class over the other and was able to generalize well on unseen test data.
- The final model demonstrates that historical launch data can be effectively used to **predict Falcon 9 landing success**.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

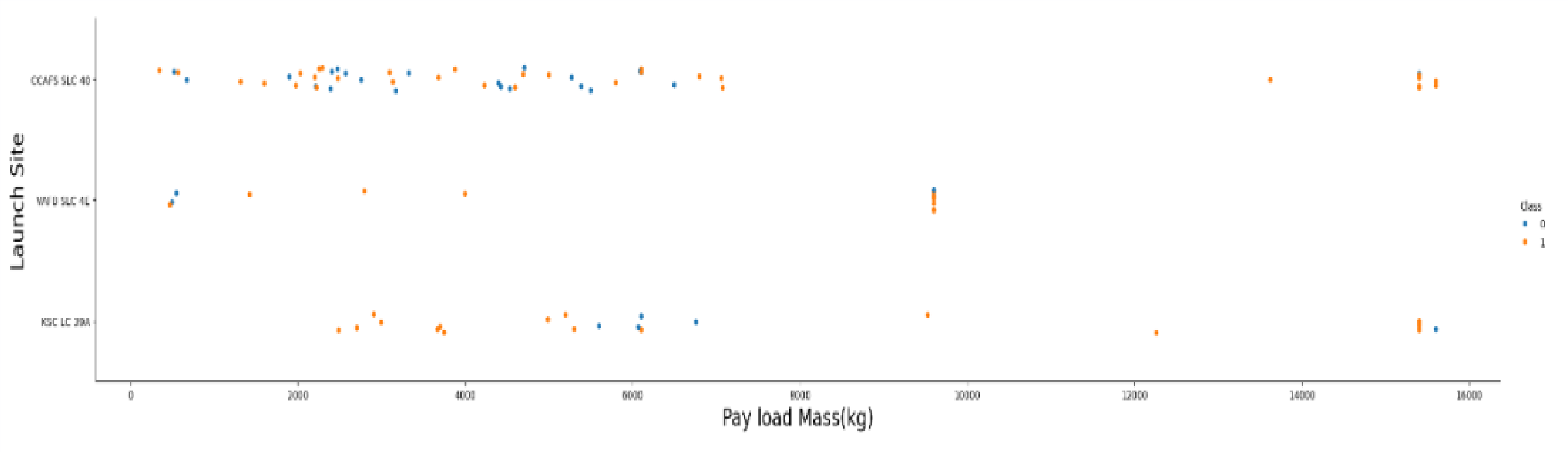
- Scatter plot of Flight Number vs. Launch Site



- Explanations: No. of launches from launch site of CCAFS SLC 40 are significantly higher than no. of launches form other sites.

Payload vs. Launch Site

- Scatter plot of Payload vs. Launch Site

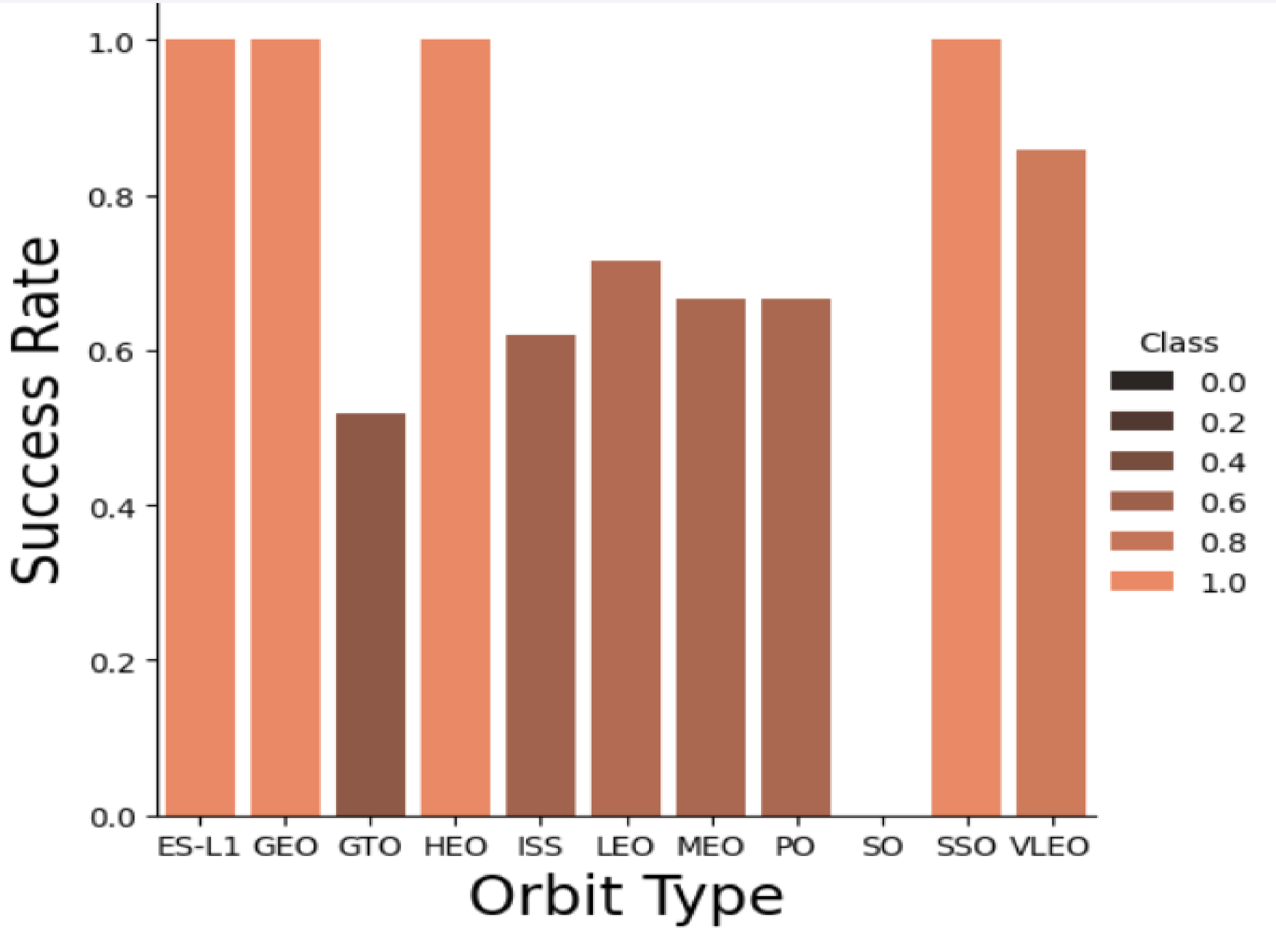


Explanations: The majority of IPay Loads with lov Mass have been launched from CCAFS SLC 40.

Success Rate vs. Orbit Type

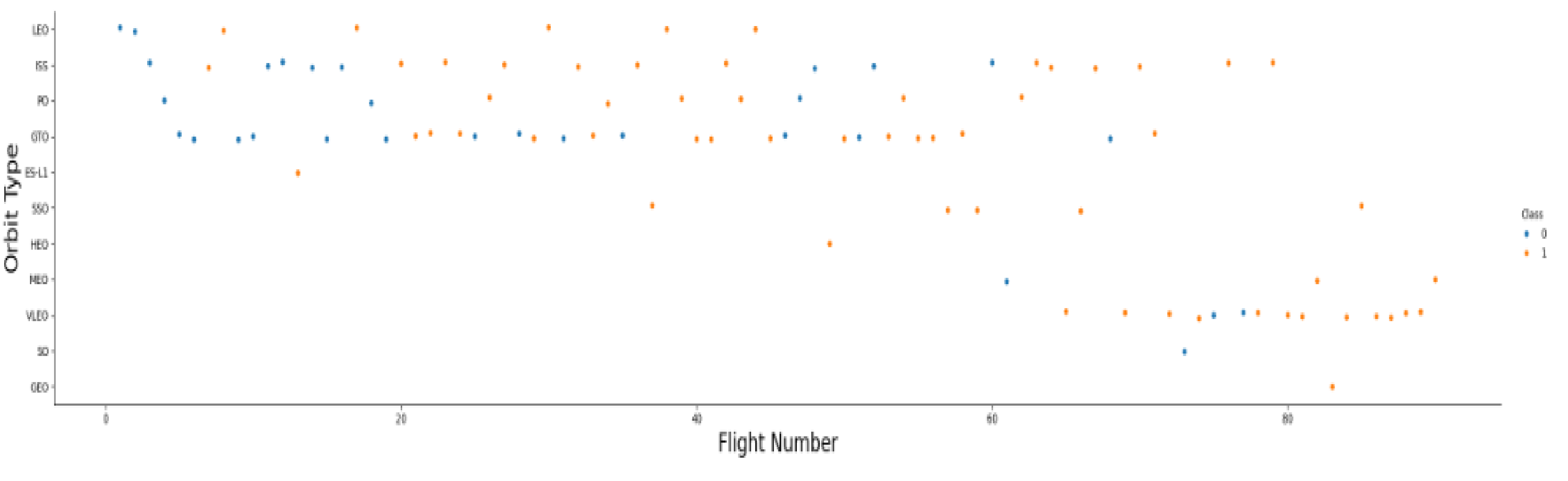
- Bar chart to depict the success rate of each orbit type

- Explanations:
- Orbit types ES-L1, GEO, HEO, and SSO have 100% success rate
- SO is having success rate of 0%



Flight Number vs. Orbit Type

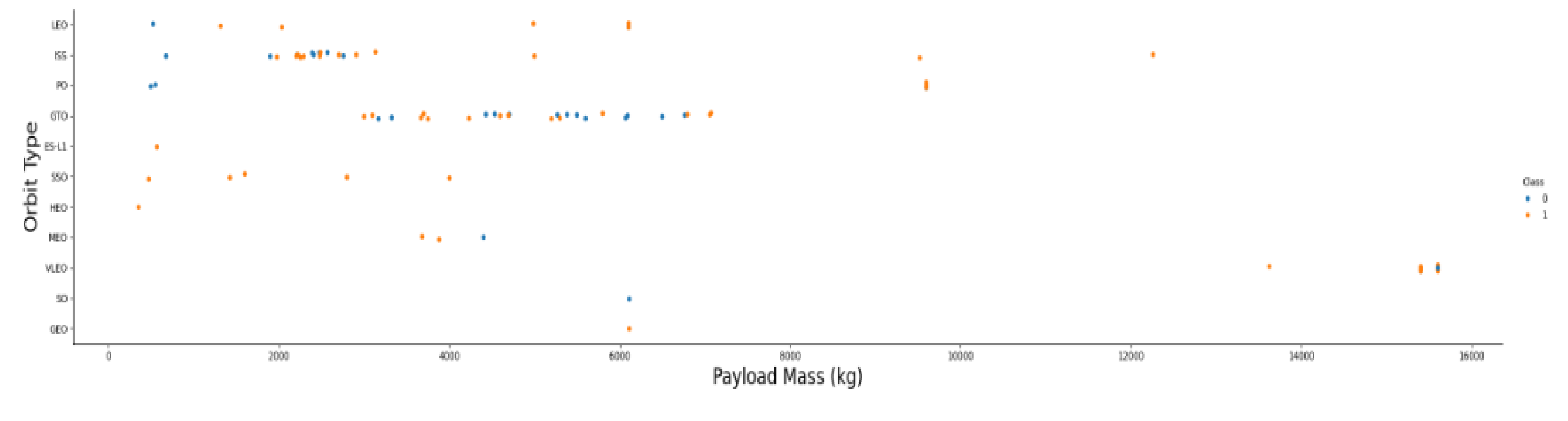
- Scatter point of Flight number vs. Orbit type



- Explanations: This plot shows, shifting of trend towards VLEO launches in recent years.

Payload vs. Orbit Type

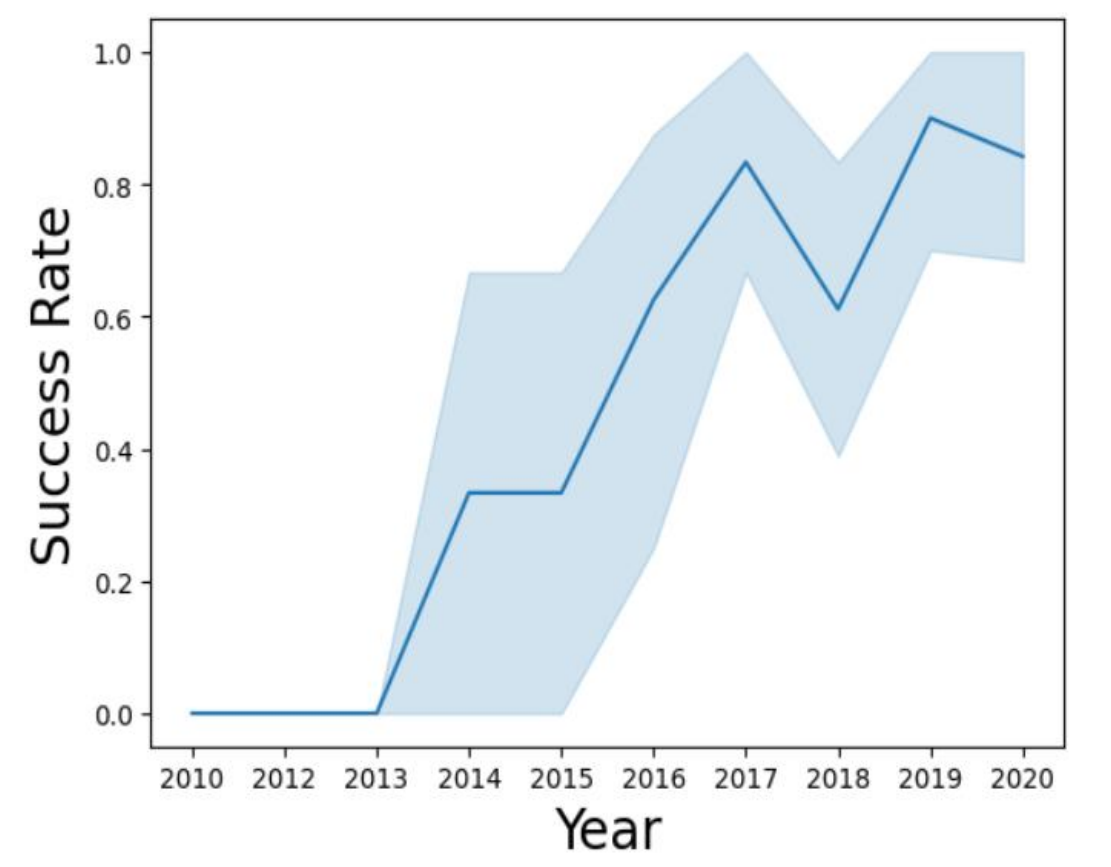
- Show a scatter point of payload vs. orbit type



- Explanations: Few obvious correlations are 1. ISS orbit and Payload at the range around 2000 Kg,   and 2. GTO orbit and the range of 4000-8000 Kg.

Launch Success Yearly Trend

- Line chart of yearly average Launch success rate
- Explanations:
- Since the year 2013, a significant increased in Launch success rate, except for year 2018.
- This is potentially due to advance in technology and expertise.



All Launch Site Names

- As Shown in the screen shot, there are 4 unique launch sites.
- For this “DISTINCT” key word was used in SQL query with select operation.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Explanation: To restrict no. of rows to 5, the LIMIT is used in SQL query

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql select sum(PAYLOAD_MASS__KG_) from SPACEXTABLE where Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

sum(PAYLOAD_MASS__KG_)
45596

- The total payload carried by boosters from NASA is shown above.
- By using SUM(), the total mass is evaluated to 45596 Kg

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%sql select avg(PAYLOAD_MASS__KG_) from SPACEXTABLE where Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

avg(PAYLOAD_MASS__KG_)
2928.4

- The average payload mass carried by booster version F9 v1.1 is evaluated to 2928.4 kg, by using sql avg().

First Successful Ground Landing Date

```
%sql select min(Date) from SPACEXTABLE where Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

min(Date)
2015-12-22

- On 22nd December 2015, the first succesful landing outcome in ground pad was acheived.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql select Booster_Version from SPACEXTABLE where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_M
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Ther are total boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000, as ahown in above query result.

Total Number of Successful and Failure Mission Outcomes

```
%sql select Mission_Outcome, count(*) from SPACEXTABLE group by Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	count(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- The total number of successful mission outcome = 99
- The totsl number of failure mission outcomes = 01

Boosters Carried Maximum Payload

- List of the names of the booster which have carried the maximum payload mass are shown here.
- Use of subquery with a suitable aggregate function in SQL query

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

```
%sql select substr(Date, 6, 2) as Month, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTABLE where substr(Date, (
```

```
* sqlite:///my_data1.db
one.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- The records of year 2015, with the month number(January and April), failure landing_outcomes in drone ship ,booster versions, launch_site, are shown here.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql select Landing_Outcome, count(*) as Count from SPACEXTABLE where Date between '2010-06-04' and '2017-03-20' group by La
```

```
 * sqlite:///my_data1.db
Done.
```

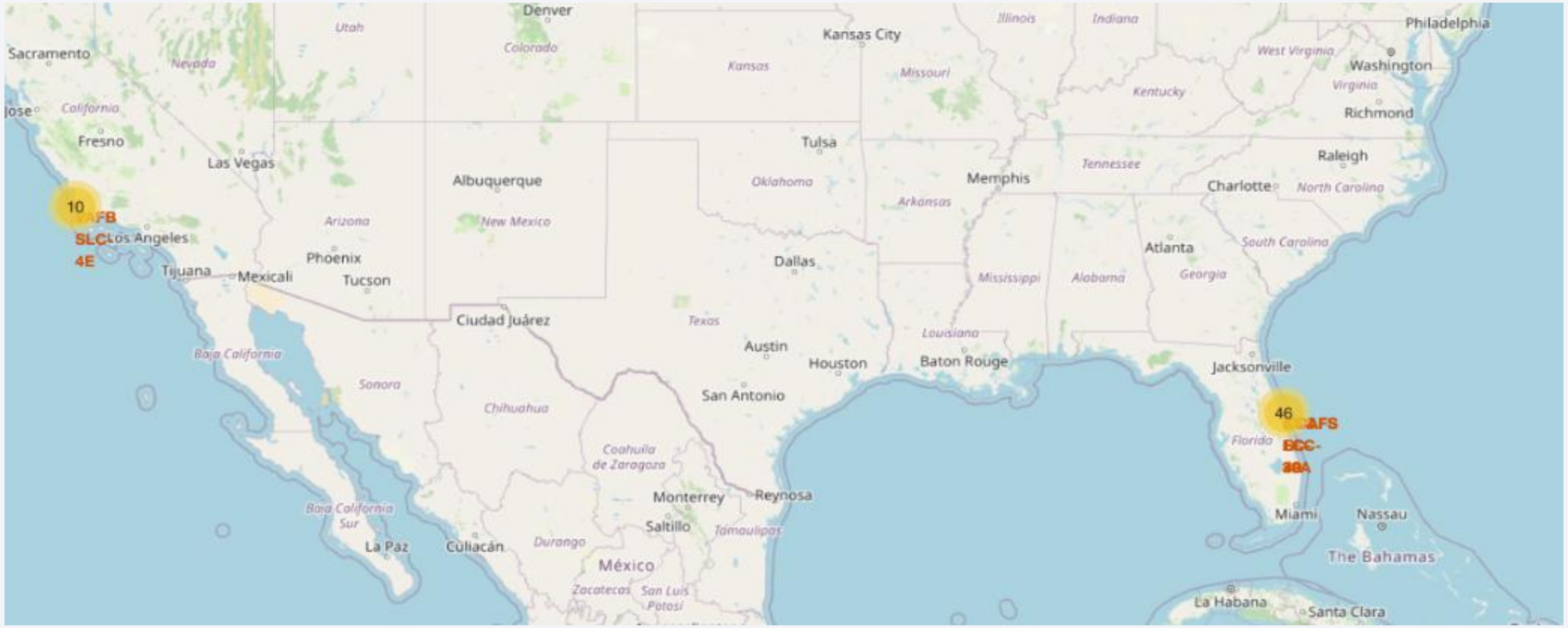
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) be tween the date 2010-06-04 and 2017-03-20, in descending order

Section 3

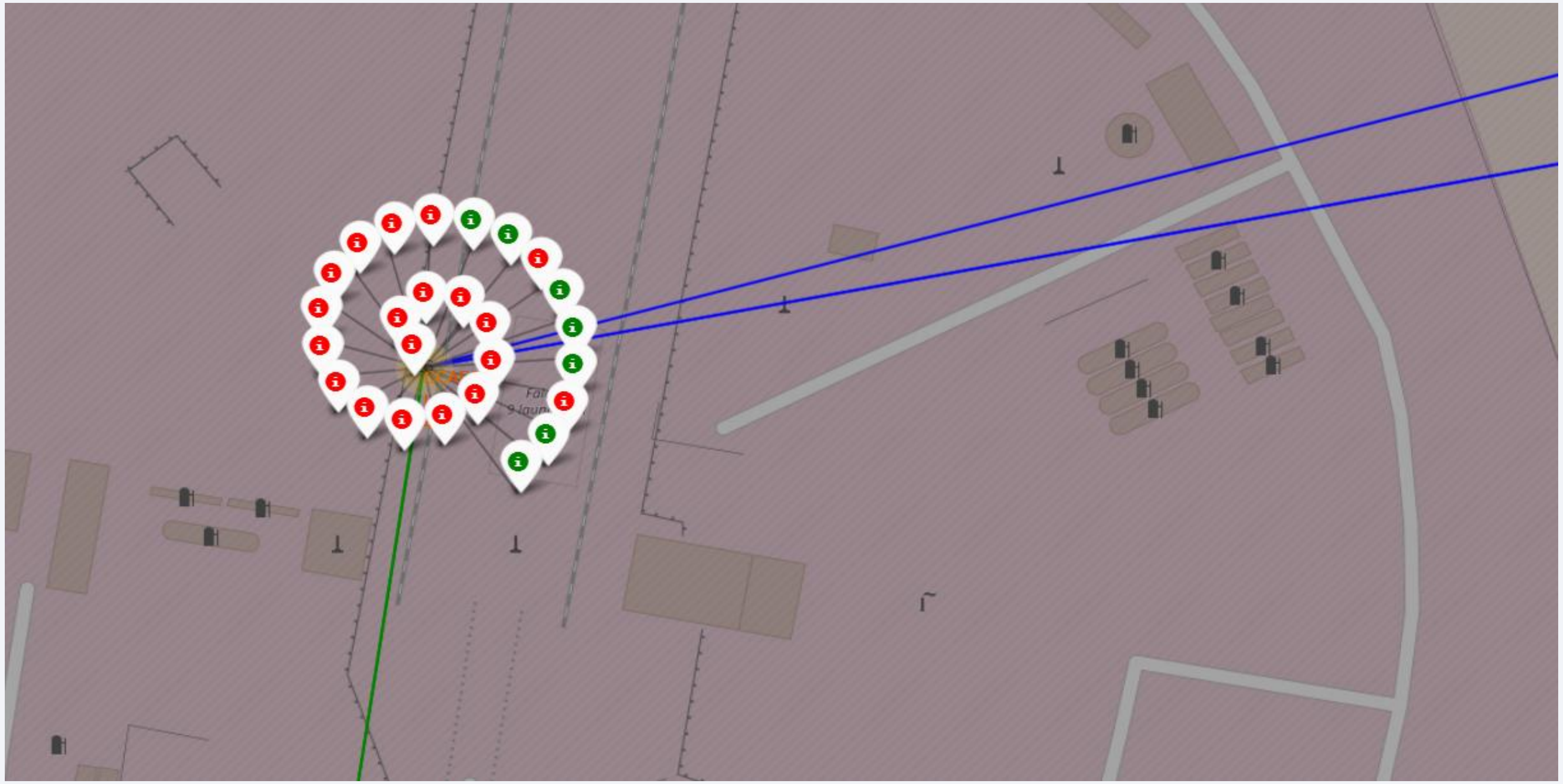
Launch Sites Proximities Analysis

Launch sites' location on a global map



- All launch sites' location markers on a global map are shown as above

The color-labeled launch outcomes on the map



The color-labeled launch outcomes on the map is shown, where the success launches for each site on the map is depicted with green color and failed launches with red color.

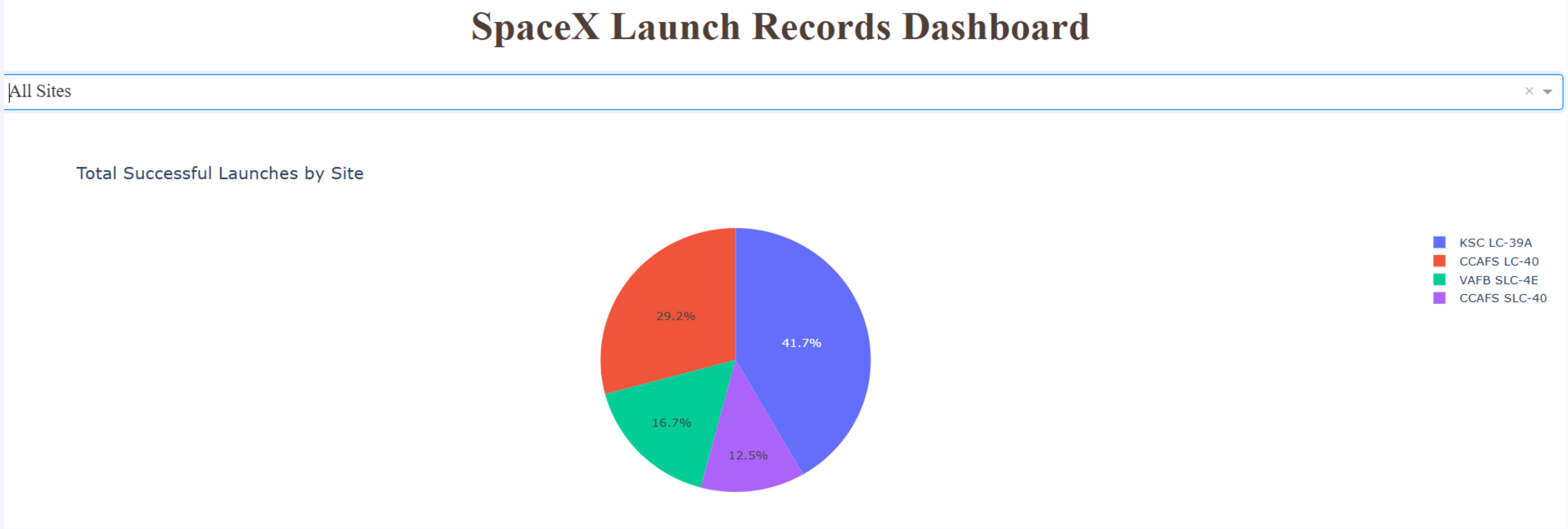
CCAFS Launch site and its proximities

- Selected launch site CCAFS to its proximities such as railway, highway, coastline, with distance calculated and displayed

Section 4

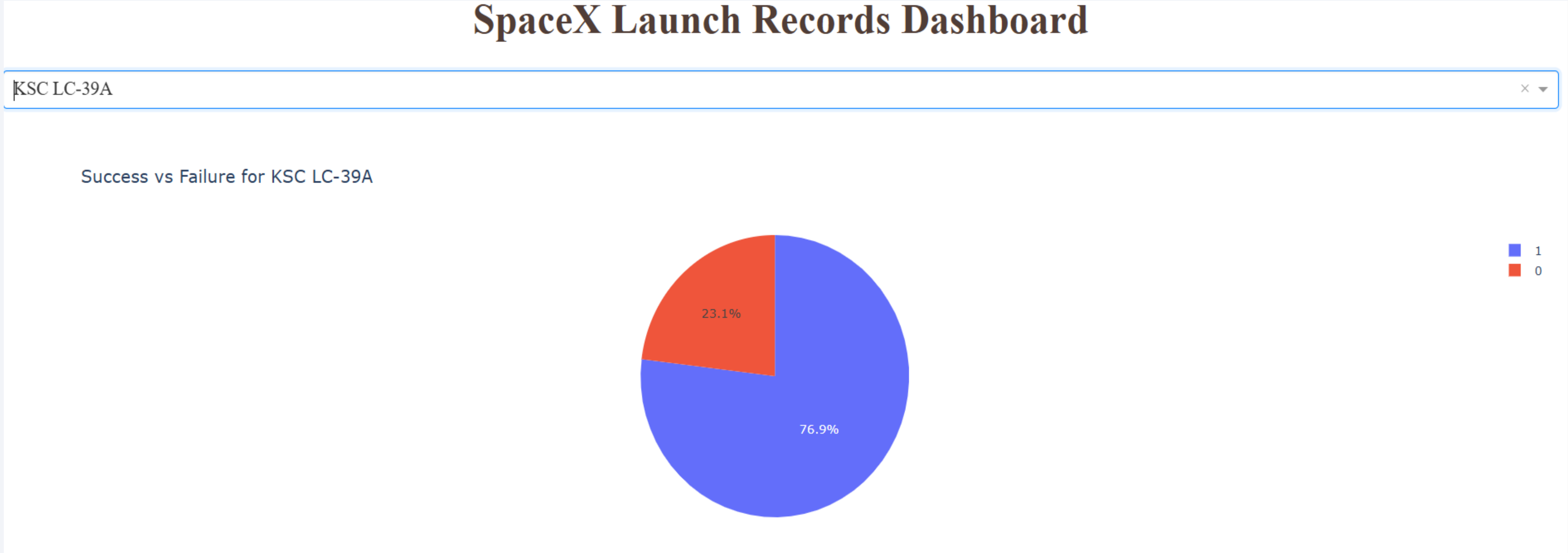
Build a Dashboard with Plotly Dash

Launch success count for all sites



- This pie chart shows the distribution of **successful SpaceX launches by all the launch site.**
- **KSC LC-39A having the highest share of successful launches**.
- Here, dropdown allows interactive filtering by launch site, helping users quickly compare **site-wise launch performance**.

Launch site with highest launch success ratio



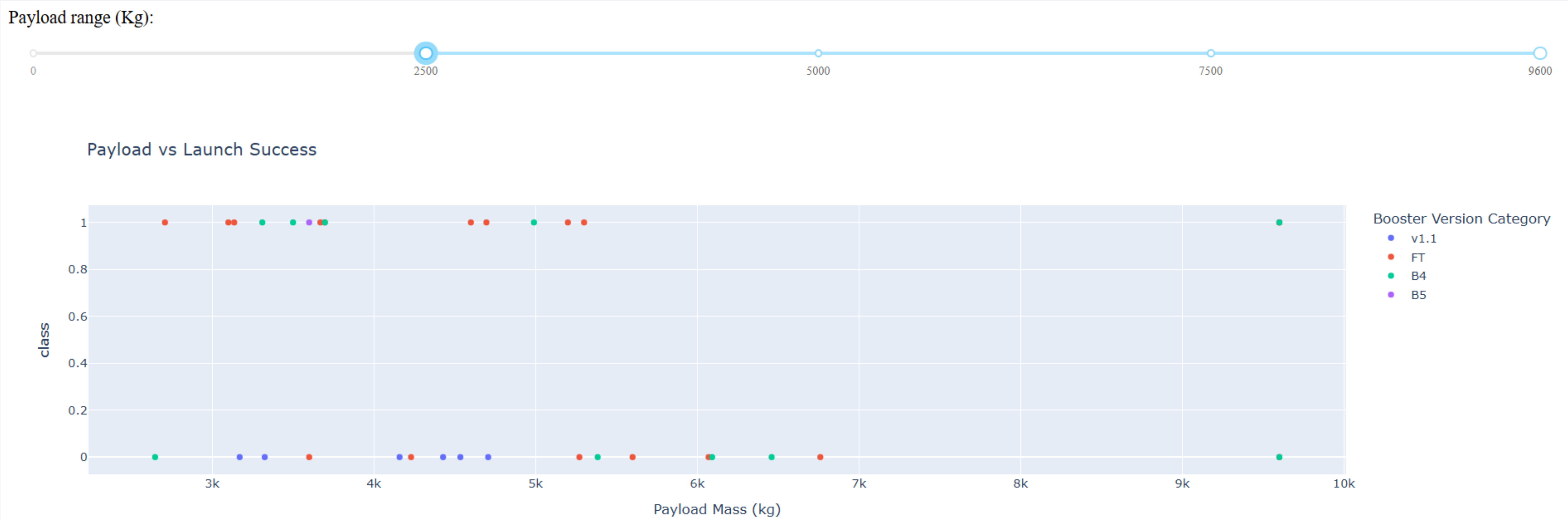
- This pie chart is showing the **success vs failure distribution** of launches at site **KSC LC-39A**.
- About **76.9% of launches were successful**, indicating a **high reliability** of the KSC LC-39A launch site than all other launch sites.

Payload vs. Launch Outcome for various range

- To analyze Payload vs. Launch Outcome scatter plot for different range, scatter plots are used.

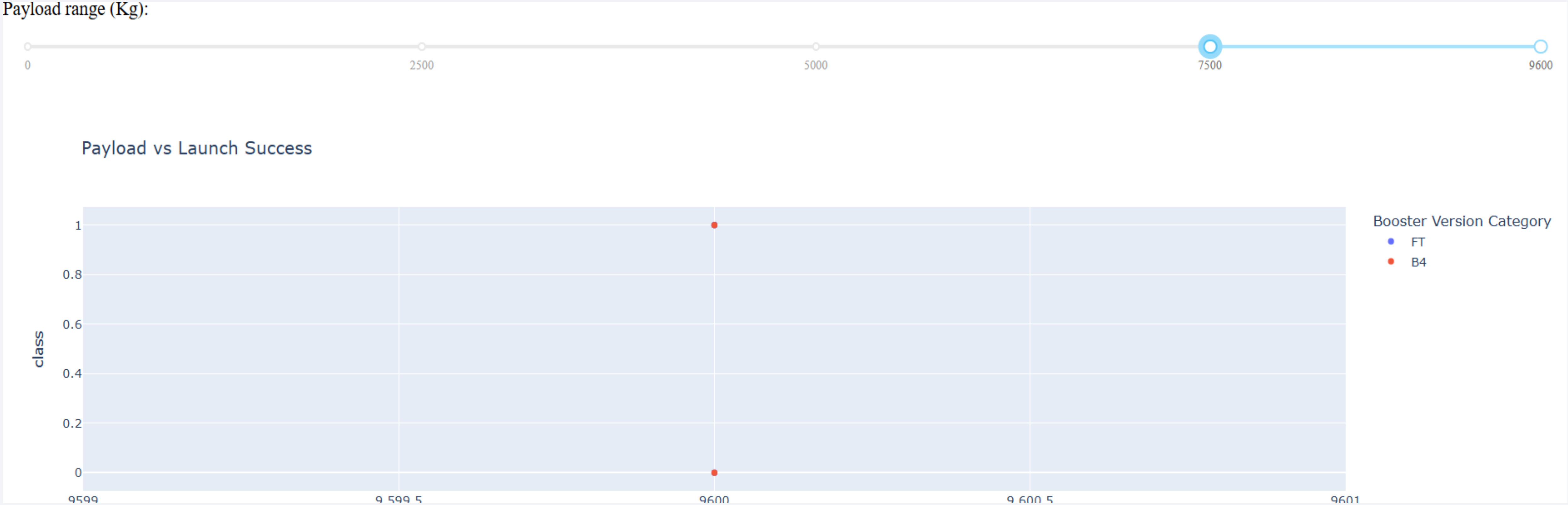
Payload vs. Launch Outcome for various range

The pevious and below scatter plots show that **lower to mid-range payload masses generally have higher success rates**, while failures are more frequent at **higher payload ranges.**



Payload vs. Launch Outcome for various range

This scatter plot indicates that increasing payload mass increases mission risk, especially above 7500 Kg.

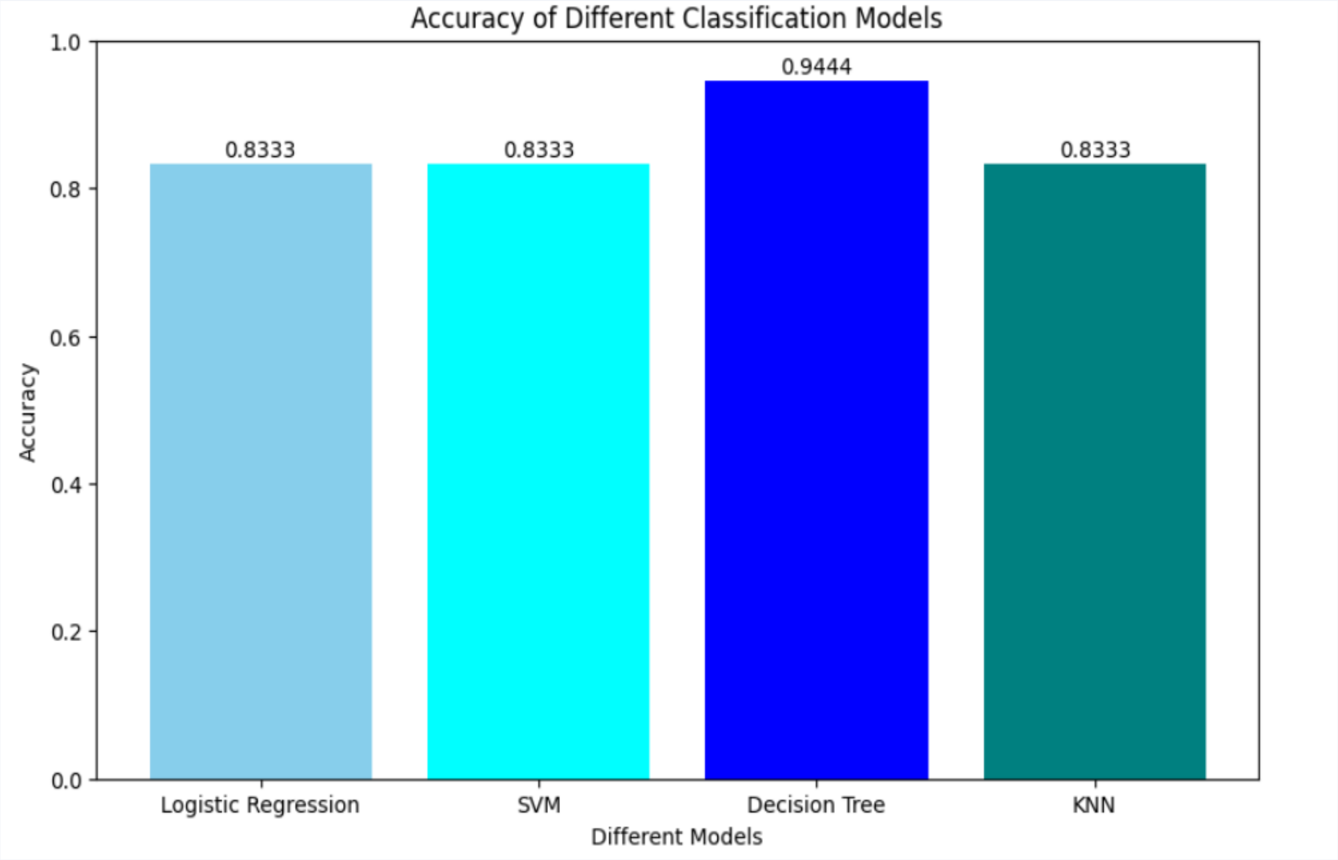


Section 5

Predictive Analysis (Classification)

Classification Accuracy

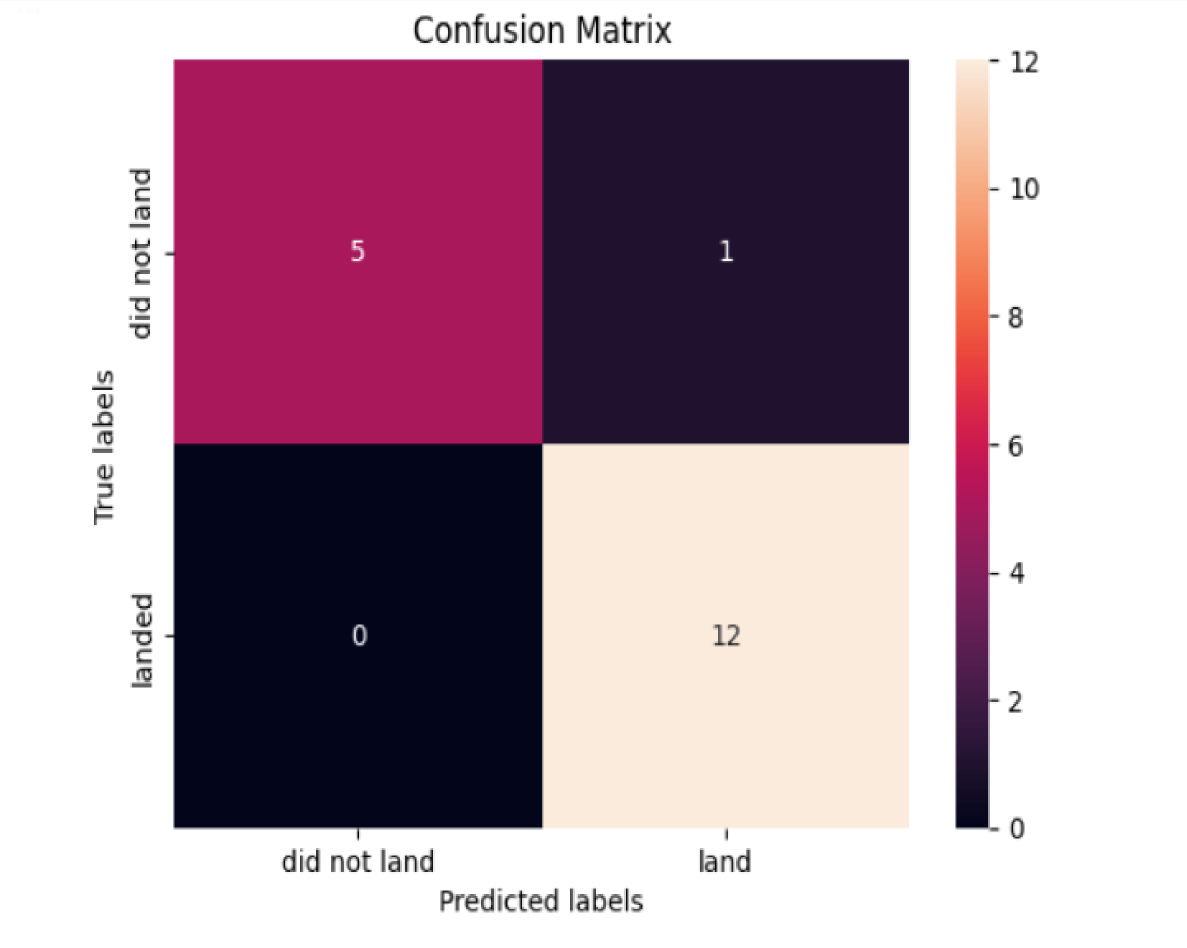
- The model accuracy for all built classification models is shown given bar chart.
- All Model performance is compared. the Decision Tree model achieved the best overall performance based on cross-validation results with highest classification accuracy of 0.9444.



Confusion Matrix

- Decision Tree Confusion matrix shows:

1. **High number of true positives and true negatives**, indicating that the model accurately classified both success and failure cases.
2. Relatively **low number of false predictions** suggests that the Decision Tree did not significantly favor one class over the other.
3. This balanced performance explains the model’s strong accuracy and reliability in predicting Falcon 9 landing outcomes.



Conclusions

- In this project, data science and machine learning techniques were successfully applied to predict the **landing success of the SpaceX Falcon 9 first stage**.
- Data collected from the SpaceX REST API and by web scraping on Wikipedia.
- Data was cleaned, analyzed, and explored using SQL, visualization, and interactive analytics tools.
- The analysis indicated that **payload mass, orbit type, and launch site** are most important in determining landing outcomes.
- The **Decision Tree model** achieved the highest accuracy because it effectively captured the **non-linear relationships** between key features such as payload mass, orbit type, and launch site.

Conclusions

- The results demonstrate that historical launch data can be effectively used to **predict landing success**, supporting better **launch cost estimation and competitive decision-making**.
- Low weighted payloads perform better than the heavier payloads.
- KSC LC 39A had the most successful launches from all the sites.
- Orbit GEO, HEO, SSO, ES L1 has the best Success Rate.

Appendix

- Grateful to all IBM Mentors for well curated content with decent presentation.

Thank you!